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Construction and analysis of corporate greenwashing index: a deep learning approach

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Abstract

Environmental information disclosure serves as a critical indicator of corporate social responsibility. While the quantity of disclosures in this area continues to rise, strategic disclosure behaviors, particularly greenwashing, are increasingly evident. This covert form of greenwashing significantly obstructs the overall advancement of ecological civilization; thus, identifying and preventing corporate greenwashing presents a key challenge in contemporary research. This study utilizes the corporate social responsibility reports of A-share companies from 2008 to 2023 as the research sample. A corporate greenwashing index is constructed using the deep learning-based MacBERT model, which demonstrates bidirectional processing capabilities. This model effectively considers contextual factors to mitigate ambiguities and extracts fine-grained information at the sentence level. The constructed corporate greenwashing index can capture the quality of corporate environmental information disclosure through text structural features, enriching the methodologies for researching corporate information disclosure and environmental behaviors and possessing significant practical application value. This research not only provides a solid foundation for subsequent exploration but also enhances the quality of corporate environmental information disclosure, assists enterprises in establishing sustainable business models, and comprehensively promotes the construction of ecological civilization.

Keywords: Deep Learning; Greenwashing Index; Corporate Social Responsibility; Environmental Information Disclosure

1 Introduction

The intensification of the global ecological crisis has brought corporate social responsibility (CSR) issues to the forefront of public concern. As primary economic entities, corporate operations have been identified as significant contributors to environmental degradation through their production activities. Empirical evidence demonstrates that over 80% of environmental pollution in China arises from corporate production and operational activities [1, 2]. As the main drivers of economic development, corporations play a crucial role in reducing carbon emissions, utilizing resources responsibly, and innovating in environmental technologies [3–5]. Corporate participation in ecological civilization construction is not only a reflection of legal obligations and social responsibilities but also an essential choice for maintaining legitimacy and achieving long-term sustainable development [6]. As the concept of sustainable development deepens, stakeholders' expectations for corporations to fulfill their social responsibilities continue to rise. Environmental information disclosure has become a significant channel for companies to respond to societal concerns and demonstrate accountability. The disclosure paradigm of corporate social responsibility (CSR) in China exhibits a unique form that lies between the voluntary disclosure model in the United States and the mandatory audit framework of the upcoming Corporate Sustainability Reporting Directive (CSRD) in the European Union. Compared with the United States, which primarily relies on the principle of voluntary disclosure by enterprises, and the European Union, which tends to adopt a mandatory audit mechanism through the CSRD, China has adopted a composite strategy that integrates "semi-mandatory" policies with market-oriented incentives when constructing its regulatory framework [6, 7]. The institutionalization of CSR disclosure commenced in 2006 when the Shenzhen Stock Exchange issued the Social Responsibility Guidelines for Listed Companies, marking the formal institutionalization of CSR disclosure practices. Subsequent implementation has demonstrated a sustained growth trajectory in CSR reporting adoption. By 2022, the number of listed companies disclosing CSR reports reached 1675, representing 32.99% of all listed entities and reflecting a 25.42% year-on-year increase. Despite the rapid expansion of environmental information disclosure, the quality of these disclosures has not kept pace with the quantity. Some corporations manipulate the content of their disclosures, exhibiting varying degrees of "greenwashing," wherein they selectively disclose favorable environmental information while intentionally concealing negative environmental performance. They may even employ ambiguous language to obscure a lack of genuine environmental action, engaging in opportunistic behavior [8, 9].

Currently, research on greenwashing is rapidly advancing [10]. Existing studies reveal various forms of greenwashing and the factors that contribute to its occurrence. Many scholars conduct multi-dimensional analyses of the consequences of corporate greenwashing from the stakeholders' perspective. The findings indicate that external factors influencing greenwashing include government regulations and enforcement [11] as well as stakeholder pressure [12]. Internal factors comprise company size [9], industry characteristics [13], financial performance [14, 15], and internal organizational capacity [16, 17]. Furthermore, current measurement methods for corporate greenwashing primarily involve setting dummy variables, designing surveys, conducting experiments, and constructing greenwashing indices [18, 19]. However, these methods have notable limitations, as they heavily rely on manual input and exhibit a degree of subjectivity and uncertainty.

With the rise of artificial intelligence and rapid advancements in natural language processing (NLP), text mining can extract information from unstructured data. The application of large language models represents a significant evolution, capturing information from entire sentences rather than merely keywords [20]. The application of deep learning models in greenwashing identification offers a new approach for recognizing corporate greenwashing behavior. The greenwashing indicators constructed based on deep learning methods not only improve the accuracy of identifying greenwashing behaviors at the textual level but also reduce reliance on manual judgment. Moreover, this method demonstrates excellent transferability and generalizability, facilitating the acquisition of large samples for further in-depth research.

The issue of greenwashing has attracted widespread attention globally, and countries are actively addressing this challenge. As the largest developing country in the world, China has a vast number of enterprises with widespread distribution. Therefore, this paper specifically focuses on Chinese listed companies, utilizing the corporate social responsibility reports of Chinese A-share listed companies from 2008 to 2023 as the research sample to construct a corporate greenwashing index through deep learning methods. This not only enriches the theoretical and methodological framework of international greenwashing research but also provides strong support for the regulation of environmental information disclosure by Chinese enterprises, thereby promoting the comprehensive development of ecological civilization construction in China.

The primary contributions of this study are summarized as follows: Firstly, this research introduces a deep learning approach to construct corporate greenwashing indicators, expanding innovative application pathways for diverse machine learning models in social science research. Secondly, the deep learning-based greenwashing indicator enables granular extraction of corporate environmental information. Unlike conventional dictionary-based methods prevalent in existing studies, the implementation of the MacBERT model incorporates contextual analysis to disambiguate terms and establish precise sentence-level greenwashing indicators [21], thereby improving measurement accuracy and precision. Thirdly, the corporate greenwashing index (GWL) based on deep learning demonstrates enhanced capability in identifying "formally compliant but substantively vacuous" greenwashing behaviors (e.g., keyword stacking and ambiguous logic) through comprehensive analysis of textual structure and semantic features. This represents a significant advancement over traditional methodologies dependent on simple indicators (e.g., word counts or dictionary-based methods), which remain vulnerable to strategic corporate disclosure practices. Fourthly, the introduction of text analysis-based deep learning models in the construction of greenwashing indicators improves data processing efficiency and increases data coverage, making the research findings more representative and universally applicable. Ultimately, the precise identification of corporate greenwashing through the MacBERT model enriches the research methods for studying corporate information disclosure and environmental behavior, promotes the transparency of corporate environmental information and the improvement of disclosure quality, and facilitates the establishment of sustainable business models by corporations.

The remaining sections of this study are organized as follows: Sect. 2 reviews the research progress on corporate greenwashing. Section 3 introduces the data sources and discusses the methodology for constructing the corporate greenwashing index. Section 4 validates the greenwashing index. Section 5 provides an analysis of the greenwashing index. Finally, Sect. 6 summarizes the significance of constructing the greenwashing index and explores its potential for various applications. Research framework is shown in Fig. 1.

2 Research progress on corporate greenwashing

2.1 Research background on corporate greenwashing

The term "greenwashing" is a combination of "whitewashing" and "green," referring to misleading environmental disclosures by corporations. The concept of greenwashing was first introduced in 1986 by American environmentalist Jay Westerveld, but relevant research only began to increase significantly around 2011 [10]. With the rise of the ESG

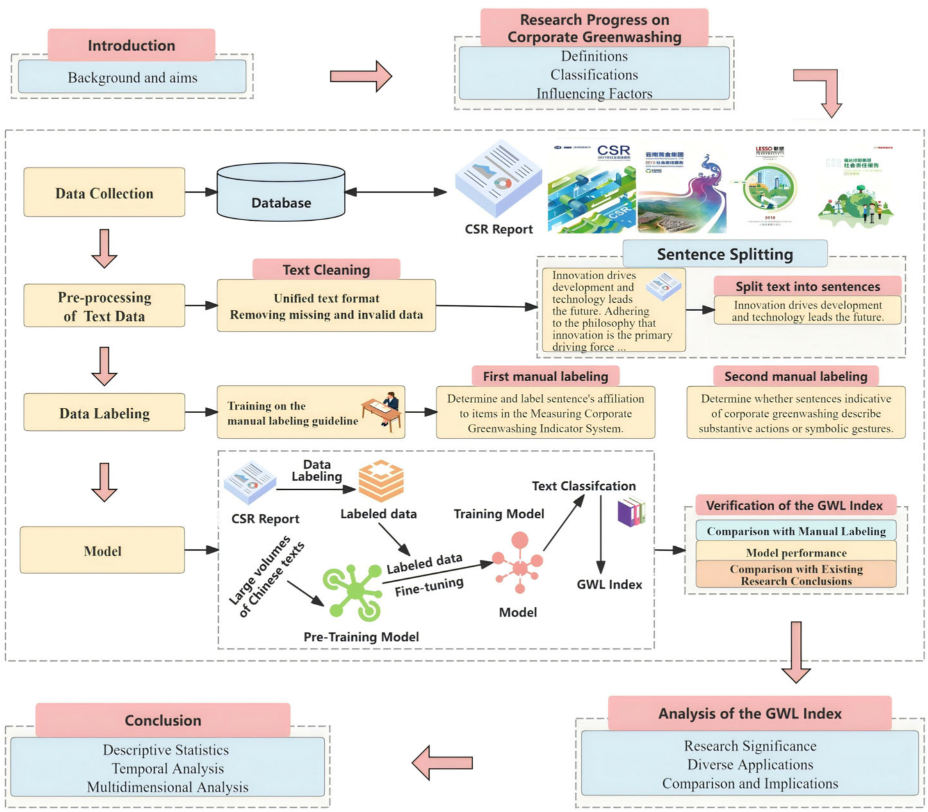

Figure 1 The framework of this study. This figure presents the comprehensive framework of this study. It delineates a meticulous journey, starting from the Introduction and proceeding through a rigorous Literature Review. The core of this framework highlights the application of the MacBERT model in constructing the GWL indicators. Finally, it includes analysis and a comprehensive summary of this study, thereby providing a clear and coherent narrative of the study's logical progression

(Environmental, Social, and Governance) concept, research on greenwashing in China has gained considerable attention across various fields, including corporate management, financial investment, and academic study, yet it remains in its early stages [22].

Human society has undergone three industrial revolutions, which, while rapidly advancing productivity, have also led to the over-exploitation of natural resources and significant ecological damage. The ongoing ecological crisis necessitates a transformation of the global economy from a traditional, extensive growth model to one focused on high-quality, sustainable development. However, during this transition, some companies fail to balance profit-seeking with social responsibility. To address increasingly stringent regulatory requirements and stakeholder pressures, these companies adopt greenwashing strategies, responding to social responsibility superficially while fundamentally opposing it [23].

Greenwashing was initially regarded as a marketing tactic employed by corporations, and subsequent in-depth research has expanded its scope and research domains. Netto et al. [24] provided three definitions of greenwashing: Greenwashing as selective disclosure; Greenwashing as decoupling; and Greenwashing based on the context of legitimacy (Signaling and corporate legitimacy theory). The first two definitions describe specific methods or forms of greenwashing, while the definition based on legitimacy focuses more on the context or motivation behind greenwashing. With the continuous improvement of information mining capabilities, the academic community has gradually broadened the research perspective on greenwashing to the level of corporate information disclosure. In the process of corporate information disclosure, greenwashing behavior often manifests as a strategic information disclosure practice. In light of this, this study aims to identify greenwashing behavior by conducting an in-depth analysis of the structure of corporate environmental information disclosure. Based on this, this study draws on the research findings of Netto et al. [24] and categorizes corporate greenwashing behavior into two types: selective disclosure and expressive manipulation. Selective disclosure refers to the selective reporting of positive environmental matters; expressive manipulation, on the other hand, measures the extent of symbolic disclosure, specifically achieving the goal of beautifying the corporate image through strategic representation. This significant gap between a company's actual environmental performance and its environmental information disclosure is, in fact, a form of decoupling in environmental information disclosure.

Companies may selectively disclose certain genuine green information to obscure or hide negative aspects, often emphasizing a limited set of real environmental practices without disclosing critical metrics, thus preventing external stakeholders from obtaining comprehensive information. Expressive manipulation involves disclosing environmental information symbolically rather than substantively, exaggerating environmental performance, making false commitments and plans, and presenting projects that appear comprehensive but lack corresponding actionable efforts. In the short term, greenwashing may help companies reduce operational costs and achieve higher profits while maintaining a positive reputation and brand image, thereby enhancing their competitive advantage in the industry and winning stakeholder support [25]. Additionally, greenwashing can lead to lower financing costs and even enable companies to receive green subsidies. Therefore, in the absence of a fully developed legal framework, many companies resort to these profit-driven, short-sighted behaviors to evade their environmental responsibilities.

2.2 Existing research on corporate greenwashing

Although research related to greenwashing has begun to show an upward trend since 2011, a review of existing research findings reveals that a comprehensive and in-depth research system has already been established. From the perspective of the evolution of conceptual connotations, academic understanding has undergone a paradigm shift from a narrow marketing strategy to a multi-dimensional governance issue. Early studies often defined greenwashing as false environmental claims made by companies in their marketing communications, which was later expanded to encompass hypocritical strategies associated with a wide range of social responsibility issues [23]. Subsequent studies define greenwashing from a comprehensive perspective, arguing that discussions should include environmental, social, and economic dimensions, thereby framing greenwashing as a collective accusation against companies for disseminating misleading environmental information [22, 25].

Additionally, in terms of typological construction, existing research has developed a multi-dimensional classification system [10, 22, 26]. Based on the level at which it occurs, greenwashing is categorized into corporate-level and product-level greenwashing [16]. According to the manner of expression, it is divided into claims-based and execution-based greenwashing. Claims-based greenwashing refers to misleading environmental claims made through textual assertions in advertising [27], while execution-based greenwashing involves the use of natural elements in advertisements without direct verbal expression to embellish the company's or product's green image [28]. Further distinctions include whether the company acts to distract attention or to decouple from its actions [26], and whether greenwashing occurs through the supply chain, leading to classifications of direct, indirect, and substitute greenwashing [22].

This study constructs a framework for categorizing greenwashing behaviors based on stakeholder characteristics by integrating multi-dimensional classification criteria. Enterprises have developed differentiated greenwashing strategy systems based on the regulatory attributes (mandatory/voluntary) of information disclosure media and the hierarchical design of content formats (data, text, images, etc.). Specifically, these behaviors can be systematically categorized into three types of greenwashing: institutional greenwashing targeting authoritative audiences (governments/investors), which primarily relies on organizational legitimacy theory and exploits information asymmetry, emphasizing institutional packaging and information manipulation. Typical strategies include enterprises using selective disclosure tactics to present a compliant image to policymakers to avoid regulatory risks [16] and systematically concealing negative environmental information. Sensory greenwashing targeting perceptive audiences (consumers/the public), which mainly leverages cognitive biases for visual misdirection and emotional marketing. This manifests as the use of ecological imagery montages in advertising communications, such as the "green slogans or natural images" combination identified by Lyon and Maxwell [27] to create a visual association with environmental protection. Parguel et al. [28] noted that the use of sensory elements like colors (blue/green) and sounds (natural sounds) in advertisements to imply environmental attributes constitutes greenwashing behavior. Normative greenwashing involving relational audiences (supply chains/industries), which primarily establishes superficial compliance through normative documents or alliance labels. For example, Berrone et al. [29] pointed out that heavily polluting industries mask actual environmental performance gaps by formulating industry green standards. Pizzetti et al. [22] found that enterprises engage in passive greenwashing due to the green endorsement behaviors of intermediaries upstream in the supply chain.

Corporate greenwashing is primarily driven by external pressures and the company's internal characteristics [9, 12, 14]. External factors include pressures from government entities, stakeholders and consumers. Market demand for green practices incentivizes companies to engage in greenwashing to enhance their reputations [30]. Information asymmetries between consumers and companies, coupled with inadequate governmental oversight, create opportunities for greenwashing [12, 31]. Internal factors involve company size, organizational capacity, and financial or non-financial performance. Larger or growth-oriented companies are more inclined to greenwash in order to meet stakeholder demands [16], and leadership styles can influence greenwashing behavior [17]. Moreover, companies facing intense industry competition may resort to greenwashing to establish their reputations during periods of loss [15, 32].

While greenwashing may yield short-term financial benefits, it ultimately harms corporate reputations and can lead to penalties and sanctions [16, 25], diminishing stakeholders' willingness to invest [33] and obstructing sustainable development. Once exposed, greenwashing can trigger a crisis of trust among consumers regarding a company's environmental claims, questioning its legitimacy [8]. This trust crisis may extend to skepticism about the company's genuine environmental actions, ultimately eroding consumer support [34]. Greenwashing behaviors can weaken the motivation for positive environmental actions among peer companies, leading to reduced investments in authentic environmental initiatives. If the market is persistently misled by greenwashing companies, it may create a scenario where "bad money drives out good," resulting in fewer companies willing to assume environmental responsibilities, thereby harming environmental sustainability [35]. The frequent occurrence of greenwashing may lead to public confusion and misunderstanding regarding environmental issues, even causing a loss of trust in environmental laws and regulations, which diminishes governmental credibility [16].

Corporate greenwashing hinders overall ecological progress. Understanding the motivations and impacts of greenwashing is essential for its prevention and mitigation. Research on corporate greenwashing is still in its preliminary stages, and identifying and mitigating greenwashing has become a key issue in the realm of corporate information disclosure, representing a current hot topic and challenge. Accurate identification of greenwashing behaviors not only serves as a foundation for subsequent research but also is a prerequisite for developing relevant governance measures. Existing methods for identifying greenwashing primarily focus on setting up dummy variables, designing surveys, employing experimental approaches, and constructing greenwashing indices. Studies that use dummy variables include Du[36], who determined whether a company engages in greenwashing based on its presence on a list published by Southern Weekend, and Gibson et al. [37], who argued that companies with low ESG scores despite signing the Principles for Responsible Investment (PRI) exhibit greenwashing behaviors. Numerous studies measure greenwashing levels using surveys [22, 38], often querying consumers or other stakeholders to assess the degree of corporate greenwashing [33, 39]. The construction of greenwashing indices is a critical direction in current research, aiming to scientifically quantify greenwashing behaviors. Various methods have been proposed in the literature, including the construction of selective disclosure indices through disclosure structure analysis [23, 31] and the extraction of vocabulary using Java technology [40]. Some methods assess greenwashing by comparing disclosed information with actual environmental performance. For instance, Kim & Lyon [32] suggested evaluating greenwashing by calculating the difference between disclosed fuel consumption and actual greenhouse gas reductions, while Wedari et al. [41] used changes in carbon emissions as independent variables and levels of environmental information disclosure as dependent variables, employing regression residuals to measure greenwashing levels. Other studies focus on the relationship between signaling effects and greenwashing to construct indices. For example, Uyar et al. [42] used ESG scores as independent variables and the number of environmental reports published as dependent variables to validate whether information dissemination acts as a signal or constitutes greenwashing. Yu et al. [43] developed a relative greenwashing index based on the disconnection between disclosure and performance. Finally, Ruiz Blanco et al. [13] utilized Bloomberg ESG scores to measure corporate action scores and calculate disclosure scores through content analysis, using the difference to assess greenwashing levels.

These methods provide diverse perspectives for constructing greenwashing indices, yet limitations remain. Survey design faces challenges regarding the appropriateness of item selection, and small sample sizes and high subjectivity are common limitations. Using substitute variables to measure greenwashing does not adequately represent the extent of corporate greenwashing objectively. Approaches involving vocabulary extraction through Java technology and dictionary methods significantly enhance objectivity and sample size but still have room for improvement in precision. With the rapid development of machine learning, text mining based on deep learning methods offers the possibility of constructing more scientifically robust greenwashing indices. Current research primarily focuses on constructing greenwashing indices from the perspective of word vectors, relying on keyword settings and dictionary configurations. However, dictionary-based methods cannot avoid ambiguities at the lexical level, leading to inaccuracies in semantic judgment and text classification. Traditional, context-free machine learning models fail to comprehensively and accurately extract the corporate greenwashing information embedded in texts. The emergence of large language models effectively addresses the challenges faced in current text mining. BERT is a deep learning-based natural language processing (NLP) model and one of the most notable models in the field of NLP. It features advanced pre-training techniques and a model structure capable of effectively handling various NLP tasks. One of BERT's distinguishing characteristics is its bidirectional processing capability, allowing it to consider the context of all words in a sentence simultaneously, rather than just preceding or following words. This bidirectionality enables BERT to excel in many NLP tasks, especially in text mining. The application of the BERT model not only significantly increases sample capacity but also accounts for context to avoid ambiguities, thereby allowing for more precise construction of greenwashing indices at the sentence level. BERT represents a milestone in natural language processing, and its application in constructing corporate greenwashing indices provides fertile ground for further research on greenwashing.

3 Construction methods of the corporate greenwashing index

3.1 Data collection

Since 2008, an increasing number of publicly listed companies in China have begun disclosing corporate social responsibility information, leading to an annual improvement in the quality of environmental information disclosures. This study selects the social responsibility reports of companies listed on the Shanghai and Shenzhen A-shares from 2008 to 2023 as the research sample. To ensure the representativeness of the sample and the accuracy of the data, financial and insurance companies, as well as ST and ST* companies(listed companies with delisting risk warnings) and those with missing variable data, were excluded, resulting in a final dataset of 9534 observations.

Extracting relevant information related to corporate greenwashing from social responsibility reports is a complex task, as it is closely tied to the context of words. Therefore, the use of BERT, a context-sensitive language model, is required to accomplish this intricate task. The BERT model was originally proposed in English contexts, and its effectiveness may vary when applied to Chinese contexts. Cui et al. [21] developed MacBERT, a pre-trained language model specifically for Chinese, based on BERT, which improved two pre-training tasks: masked language modeling (MLM) and next sentence prediction (NSP). Its performance significantly outperforms other Chinese pre-trained language models. Consequently, this study will utilize the MacBERT model, based on deep learning algorithms, to construct the corporate greenwashing index for Chinese-listed companies.

3.2 Data labeling

3.2.1 Theoretical basis for data labeling

Firstly, based on the definition of corporate greenwashing behavior in this paper, corporate greenwashing strategies are ultimately defined as two approaches: selective disclosure and expressive manipulation. The former refers to the selective reporting of environmental matters, while the latter involves the strategic representation to beautify the company's image. Secondly, in accordance with relevant laws, regulations, and industry standards, Huang and Chu [23]constructed an indicator system for measuring corporate greenwashing. Theoretically, companies should truthfully disclose all the detailed items included in the measuring corporate greenwashing indicator system. However, in practice, companies exhibiting greenwashing often engage in selective disclosure of their green commitments or actions to obscure their poor performance or unfulfilled promises. Therefore, this study employs the ratio of undisclosed items to the total required disclosures to measure the extent of selective disclosure.

The specific calculation formula is as follows:

$$\text{Selective Disclosure (GWLS)} = 100 \times \left(1 - \frac{\text{Number of Disclosed Items}}{\text{Total Required Items}}\right)$$

In the process of disclosing environmental performance, companies frequently employ various textual strategies, providing detailed descriptions of specific environmental activities or strategic claims while being vague about inadequately performed or unimplemented activities. Consequently, it is necessary to assess whether the environmental performance reported by the company reflects substantive actions or merely symbolic gestures. Existing research indicates that when companies disclose verifiable, hard-to-replicate information through factual statements, case studies, or quantitative descriptions, the reliability of their environmental information is high, signifying substantive action [8]. Conversely, if a company's environmental report predominantly consists of aspirational statements, qualitative disclosures, or simple repetitions of previous years' claims that presenting unverifiable and easily mimicked information, then the reliability of its environmental disclosures is low, indicating symbolic gestures [23]. This study measures the extent of expressive manipulation by employing the ratio of symbolic disclosures to total disclosed items.

The specific calculation formula is as follows:

$$\text{Expressive Manipulation (GWLE)} = 100 \times \left(\frac{\text{Number of Symbolic Disclosures}}{\text{Total Disclosed Items}}\right)$$

3.2.2 Data labeling process

1. Development of Audit Guidelines and Training

Given that the subjects of the study are listed companies, and in combination with the environmental governance and environmental protection investment information disclosed by enterprises in their social responsibility reports, this paper adopts the classification standards and calculation methods proposed by Huang and Chu [23] as the basis for constructing corporate greenwashing indicators, following the principles of rationality and practicality. The research of Huang and Chu [23] summarized the matters that should be disclosed in corporate environmental reports under an ideal model, based on relevant laws, regulations, industry standards, and guidelines. These matters were categorized into four modules: Governance and Structure, Processes and Controls, Inputs and Outputs, and Compliance and Legality, totaling 20 sub-items. This formed an indicator system for measuring corporate greenwashing, which was further applied as a manual labeling guide to ensure consistency and accuracy in the evaluation process. The details of the measuring corporate greenwashing indicator system can be found in Appendix A.

Subsequently, the labeling personnel were trained using the manual labeling guide. The labeling personnel consisted of three researchers with significant relevant knowledge backgrounds who independently labeled the data according to the guide. After multiple iterations of training, the accuracy of the labeling results was ultimately ensured.

2. Formal Manual Labeling

Data Labeling 1: During the labeling process, each sentence was independently labeled by the three researchers with significant relevant knowledge backgrounds according to the labeling guide. If a sentence pertained to any of the 20 sub-items in the measuring corporate greenwashing indicator system, it was labeled as a sentence belonging to the content of the measuring corporate greenwashing indicator system (i.e., "1"); otherwise, it was labeled as a sentence not belonging to the content of the indicator system (i.e., "0"). Only when a sentence was judged by all three researchers to belong to the content of the measuring corporate greenwashing indicator system was it finally labeled as such. Ultimately, a total of 6493 sentence samples were labeled in Data Labeling 1.

Data Labeling 2: During the labeling process, each sentence was again independently labeled by the three researchers with significant relevant knowledge backgrounds according to the labeling guide. This time, the labeling required determining whether a sentence belonging to the content of the measuring corporate greenwashing indicator system described substantive actions or symbolic gestures. Similarly, the labeling was done independently according to the guide, and only when a sentence was judged by all three researchers to describe substantive actions was it labeled as such (i.e., "1"). Conversely, sentences describing symbolic gestures were labeled accordingly (i.e., "0"). Ultimately, a total of 5956 sentence samples were labeled in Data Labeling 2. Examples of labeled samples are provided in Appendix B.

3.3 Model

This paper employs the MacBERT model based on deep learning, which comprises two steps: pre-training and fine-tuning [44]. During the pre-training phase, the MacBERT model learns contextual associations between words in large-scale unlabeled data by bidirectionally analyzing the relationships between words in their contexts. The MacBERT model has been trained on a vast amount of Chinese text, thereby mastering the linguistic rules and semantic features of Chinese. On this basis, further domain-specific pre-training is conducted to enhance its understanding and processing capabilities for Chinese environmental information text. Subsequently, in the fine-tuning step, the MacBERT parameters are further trained using a manually labeled dataset to improve the model's accuracy. The specific steps are as follows:

1. Pre-training

This study selects MacBERT [21] as the base model, which employs whole-word masking and replaced token detection, demonstrating superior performance compared to the traditional BERT model in Chinese semantic understanding tasks. Building on the

MacBERT model trained by Cui et al. [21], we further conduct domain-specific pre-training using the "Environment and Social Responsibility" content from the annual reports of listed companies to enhance the model's semantic representation capabilities in the field of environmental governance, ensuring that the model is better suited for the research questions of this study.

2. Model Fine-tuning

This study implemented two independent model fine-tuning procedures targeting the GWLS and GWLE indicators, utilizing two distinct independently labeled datasets. The detailed workflow is structured as follows:

Fine-tuning 1 (Fine-tuning for GWLS): Here, the manually labeled datasets are utilized to train the MacBERT parameters to better reflect the relevance of GWLS-related sentences (i.e., whether a sentence belongs to the Measuring corporate greenwashing indicator system). 80% of the labeled datasets are randomly selected as the training datasets, 10% as the validation set, and the remaining 10% as the testing datasets. The model iterates on the training set, adjusting internal weights and biases to minimize prediction errors. The validation set is used for tuning the optimizer hyperparameters. Finally, the model's performance is tested on the test set. During the training phase, the classification functionality is realized through the linear layer, which generates scores relative to each class label. The Softmax activation function then converts these scores into probabilities. Standard cross-entropy loss is used to optimize the training task:

$$L\left(\overrightarrow{s}, \overrightarrow{p}\right) = -\sum_{i=1}^{2} s_i log p_i$$

$\mathrm{L}\left(\overrightarrow{\mathrm{s}}, \overrightarrow{\mathrm{p}}\right)$ is the classification loss value for each sentence item. $\overrightarrow{\mathrm{s}}$ is a two-dimensional vector and represents the actual classification of the sentence. If $s_1 = 1$ and $s_2 = 0$, the sentence belongs to the measuring corporate greenwashing indicator system. If $s_1 = 0$ and $s_2 = 1$, it doesn't. $\overrightarrow{\mathrm{p}}$ is also a two-dimensional vector and represents the outputs of the model, and the i-th element p_i in $\overrightarrow{\mathrm{p}}$ indicates the probability that sentence belongs to the i-th class label.

Fine-tuning 2 (Fine-tuning for GWLE): Here, the manually labeled datasets are used to train the MacBERT parameters to better reflect the relevance of GWLE-related sentences (i.e., whether a sentence belongs to substantive actions or symbolic gestures). 80% of the labeled datasets are randomly selected as the training datasets, 10% as the validation set, and the remaining 10% as the testing datasets. The model iterates on the training set, adjusting internal weights and biases to minimize prediction errors. The validation set is used for tuning the optimizer hyperparameters. Finally, the model's performance is tested on the test set. During the training phase, the classification functionality is realized through the linear layer, which generates scores relative to each class label. The Softmax activation function then converts these scores into probabilities. Standard cross-entropy loss is used to optimize the training task:

$$L\left(\overrightarrow{E}, \overrightarrow{p}\right) = -\sum_{i=1}^{2} E_i log p_i$$

$L\left(\overrightarrow{E}, \overrightarrow{p}\right)$ is the classification loss value for each sentence item. $\overrightarrow{E}$ is a two-dimensional vector and represents the actual classification of the sentence. If $E_1 = 1$ and $E_2 = 0$, the sentence belongs to the substantive action. If $E_1 = 0$ and $E_2 = 1$, it doesn't, in other words, the sentence is categorized as belonging to symbolic gestures. $\overrightarrow{p}$ is also a two-dimensional vector and represents the outputs of the model, and the i-th element p_i in $\overrightarrow{p}$ indicates the probability that sentence belongs to the i-th class label.

3. Model Inference

Finally, the trained model is applied to all sentence samples to construct the GWLS and GWLE indicators, respectively. The degree of corporate greenwashing (GWL) is then represented by the geometric mean of the degrees of selective disclosure and expressive manipulation, calculated as follows:

$$GWL = \sqrt{GWLS \times GWLE}$$

4 Verification of the corporate greenwashing index

4.1 Comparison with manual labeling

To evaluate the consistency between manual labeling and deep learning results, four metrics were employed: Accuracy, Precision, Recall, and F1 Score. Accuracy represents the percentage of correctly predicted samples relative to the total number of samples. Precision indicates the probability that samples predicted as positive are actually positive, while Recall measures the proportion of actual positive samples correctly predicted by the model. The F1 Score provides a comprehensive evaluation by combining both Precision and Recall.

For the GWLS index constructed by the MacBERT model, an accuracy of 97.79% was achieved, indicating that 97.79% of the model's predictions were consistent with manual labeling, reflecting the model's overall excellent performance. The precision was 97.98%, signifying that 97.98% of samples predicted as positive were indeed positive, demonstrating the model's high accuracy in identifying positive instances. The recall reached 99.67%, indicating that the model identified nearly all positive instances, showcasing its strong recognition capability. The F1 score was 98.82%, indicating a balance between precision and recall that reflects the model's ability to accurately identify positive instances while maintaining broad coverage.

Similarly, for the GWLE index derived from the MacBERT model, an accuracy of 95.13% was observed, meaning that 95.13% of the predictions were consistent with manual labeling, which still indicates superior performance. The precision was 95.54%, showing that among samples predicted as positive, 95.54% were actual positives, reflecting a high level of reliability. The recall was 95.64%, demonstrating that the model correctly identified 95.64% of actual positives, thus exhibiting effectiveness in recognizing positive instances. The F1 score of 95.59% indicates a good balance between precision and recall.

Overall, the superior performance exhibited by the MacBERT model in calculating the GWLS and GWLE indices suggests a high degree of alignment between its quantitative results and those from manual labeling. This implies that the MacBERT model possesses significant accuracy and stability when identifying whether corporate environmental disclosures exhibit greenwashing behavior, effectively recognizing and predicting relevant outcomes.

Table 1 Performance comparison results of four models (MacBERT, RoBERTa, PERT, LERT) on the testing dataset. The table displays the accuracy, F1 score, precision, and recall for each model, assessed across two metrics: GWLS (Selective Disclosure) and GWLE (Expressive Manipulation). GWL denotes the degree of corporate greenwashing. The results reveal that all models achieve high accuracy, with MacBERT demonstrating superior performance in most evaluation metrics, indicating its effectiveness in identifying corporate greenwashing behavior

	MacBERT		RoBERTa		PERT		LERT	
	GWLS	GWLE	GWLS	GWLE	GWLS	GWLE	GWLS	GWLE
Accuracy	97.79%	95.13%	97.69%	94.85%	97.54%	94.85%	97.59%	94.85%
F1	98.82%	95.59%	98.76%	95.30%	98.68%	95.35%	98.71%	95.35%
Precision	97.98%	95.54%	97.82%	95.98%	97.76%	94.87%	97.71%	94.96%
Recall	99.67%	95.64%	99.72%	94.62%	99.61%	95.84%	99.72%	95.74%

4.2 Model performance and robustness

To ensure that the results are not influenced by model selection and to validate their robustness, this study introduced alternative models and conducted a comparative analysis of their results against those obtained from the MacBERT model. Specifically, the RoBERTa, PERT, and LERT models were selected as benchmark comparisons, and the performance of all four models was evaluated using the same dataset. As shown in Table 1, the MacBERT model achieved the highest accuracy and F1 score, demonstrating superior performance overall. Although the accuracy of the MacBERT model was slightly lower than that of the GWLE derived from the RoBERTa model, it still surpassed the other models. Furthermore, the recall rate of the MacBERT model was comparable to that of the other models, reflecting excellent overall performance. Thus, the MacBERT model is considered the optimal choice as the primary model.

To further validate the robustness of the models, the GWL index was recalculated using the RoBERTa, PERT, and LERT models, and the results were compared with those from the MacBERT model. Correlation analysis revealed that the correlation coefficients between the GWL indices calculated by the RoBERTa, PERT, and LERT models and those from the MacBERT model were 95.88%, 94.90%, and 94.47%, respectively. This indicates that the computation of the GWL index is not influenced by model selection, thus confirming its robustness.

4.3 Comparison with existing research conclusions

Existing studies indicate that corporate greenwashing behaviors are influenced by both internal drivers and external pressures. Internally, the motivations and management strategies of firms play a crucial role. To maximize sales revenue, companies may engage in greenwashing to cultivate a green image that attracts consumers and investors. However, this motivation often accompanies a lack of commitment to genuine green development, further exacerbating the risks of greenwashing [45]. Additionally, the greater the managerial myopia, the more pronounced the greenwashing phenomenon becomes, revealing the impact of short-term-oriented management decisions on greenwashing behaviors [46]. Externally, government policies, market conditions, societal oversight, and the rapidly evolving digital environment significantly impact corporate greenwashing behaviors. For instance, extensive attention on social media may compel companies to adopt greenwashing strategies in response to public scrutiny and pressure [47]. Conversely, digital transformation has been found to effectively mitigate greenwashing behaviors. Research indicates that for every 1% increase in digital transformation, corporate greenwashing decreases by

Table 2 Comparison of regression coefficients and significance levels for variables related to corporate greenwashing (GWL) in relation to existing empirical conclusions. The table presents three variables: Myopia (managerial myopia), Media (social media), and DIGI (digital transformation), along with their respective regression coefficients and significance levels. Results indicate that the relationships identified in this study are consistent with the existing literature, revealing positive correlations for managerial myopia and social media, and a negative correlation for digital transformation. This alignment reinforces the robustness of the findings across various studies

Variables and GWL			Regression coefficient and significance		
			Theory	This study	Consistency
1	X - Myopia	Y - GWL	Positive	Positive	Yes
2	X - Media	Y - GWL	Positive	Positive	Yes
3	X - DIGI	Y - GWL	Negative	Negative	Yes

3.42% [48–50], highlighting the importance of digital tools in enhancing corporate transparency and management efficiency. The empirical findings of this study align with existing theories, demonstrating the effectiveness of the constructed corporate greenwashing index. Detailed information can be found in Table 2.

5 Analysis of the corporate greenwashing index

5.1 Descriptive statistics

The corporate greenwashing index (GWL), constructed using the deep learning model MacBERT, encompasses a comprehensive dataset of 9534 data points. Notably, some companies have exhibited a gradual decline in their greenwashing indices, indicating a potential reduction in their greenwashing behaviors. However, a significant proportion of firms continue to sustain elevated greenwashing indices, revealing a marked polarization in corporate greenwashing practices. This phenomenon is further elucidated in Table 3, which provides a detailed analysis of the variations in greenwashing behaviors across different companies.

From 2008 to 2023, the average GWL decreased from 75.69 to 62.75, indicating an overall reduction in corporate greenwashing behaviors during this period. Notably, the standard deviation in 2023 reached 16.48, the highest among all years, suggesting increased differentiation in corporate greenwashing performance. While some firms significantly reduced their greenwashing activities, others may have maintained or even intensified these behaviors.

During the period from 2008 to 2013, the average GWL remained relatively stable, fluctuating between 75 and 71. However, starting in 2014, the average GWL began to decline significantly, especially after 2021, when it dropped to a low of 62.75. This change may reflect recent efforts by companies to reduce greenwashing in response to increasing societal and regulatory pressures. It is important to note that from 2021 to 2023, the standard deviation of GWL continued to increase, further indicating that, despite an overall reduction in greenwashing behaviors, the disparity among companies has widened. This is evidenced by some firms significantly improving their environmental transparency while others possibly continued to maintain or exacerbate their greenwashing behaviors. Additionally, the maximum GWL values remained high across all years, suggesting that certain companies consistently exhibited severe greenwashing behaviors. Conversely, the minimum GWL values experienced significant fluctuations between 2016 and 2023, with the lowest point of 19.00 occurring in 2021. This may reflect the adoption of more genuine and transparent environmental measures by some companies during this time.

Table 3 Descriptive statistics of the Degree of Corporate Greenwashing (GWL) from 2008 to 2023. This table presents annual data, including sample size (N), mean, minimum, maximum, 25th percentile (p25), median (p50), 75th percentile (p75), and standard deviation (SD). The results reveal significant trends in GWL over the specified period, illustrating notable shifts in corporate greenwashing behavior and providing insights into the evolving landscape of corporate environmental practices

Year	N	Mean	Min	Max	p25	p50	p75	SD
2008	137	75.69	40.00	97.47	65.63	77.46	87.83	14.17
2009	124	73.37	46.10	97.47	61.57	73.52	82.81	13.31
2010	382	72.41	31.62	97.47	61.58	73.03	82.46	14.14
2011	468	71.76	29.58	97.47	61.85	71.76	79.84	13.45
2012	488	71.94	38.24	97.47	60.94	71.18	81.65	14.01
2013	459	71.43	32.37	97.47	61.24	70.50	81.65	14.77
2014	487	71.71	25.50	97.47	61.24	70.71	82.46	14.55
2015	589	72.30	29.81	97.47	62.31	71.81	82.16	14.21
2016	576	71.10	35.86	97.47	60.94	70.71	80.18	14.34
2017	658	70.95	20.84	97.47	61.24	70.71	80.18	14.65
2018	720	70.50	29.05	97.47	60.94	70.33	80.05	14.60
2019	747	69.36	29.60	97.47	58.88	69.08	79.84	14.67
2020	836	68.74	28.22	97.47	57.59	68.31	79.06	15.15
2021	1045	66.32	19.00	97.47	54.77	66.14	77.46	15.40
2022	996	64.45	23.03	97.47	52.96	63.81	75.28	15.62
2023	822	62.75	24.70	97.47	50.11	62.26	73.85	16.48
Total	9534	69.11	19.00	97.47	58.31	68.76	79.84	15.21

In summary, both the median and average GWL have significantly decreased since 2021, indicating a general reduction in greenwashing behaviors among companies. However, the increasing standard deviation and consistently high maximum values imply that, while the majority of firms may be responding to societal and regulatory pressures by reducing greenwashing, some still maintain or even intensify such behaviors, leading to a more dispersed data distribution. This suggests that, despite some progress in curbing greenwashing, significant challenges remain.

5.2 Temporal analysis of corporate greenwashing index

In 2008, the Shanghai Stock Exchange issued the "Guidelines for Environmental Information Disclosure by Listed Companies" and the "Guidelines for the Preparation of Corporate Social Responsibility Reports," encouraging companies to proactively disclose environmental information. The average GWL during 2008 to 2010 remained between 75.69 and 72.41, reflecting a widespread prevalence of greenwashing behavior among companies. The proximity of the mean to the median indicates that many companies engaged in greenwashing; however, the high maximum value of 97.47 and the wide range from 19.00 to 97.47 demonstrate significant disparities in environmental performance among different firms. This reflects the initial impact of the policy framework, although the lack of mandatory regulations resulted in noticeable differences in environmental performance across companies.

In 2010, the Ministry of Environmental Protection released the "Guidelines for Environmental Information Disclosure by Listed Companies," providing more systematic guidance for corporate environmental disclosures. Between 2011 and 2017, the GWL average fluctuated around 71, showing no significant improvement, which indicates that the initial implementation of policies had limited impact on overall corporate greenwashing behavior. The relative stability of the standard deviation suggests a consistent performance among companies regarding greenwashing, as the influence of policy began to emerge but had yet to reach full effect.

In 2017, the Ministry of Environmental Protection and the China Securities Regulatory Commission further enhanced the requirements for compliance and transparency in corporate environmental disclosures. With the gradual implementation of regulations such as the "Measures for the Administration of Legal Disclosure of Enterprise Environmental Information," companies faced stricter disclosure requirements. From 2017 onwards, the average GWL began to decline significantly, dropping from 70.95 in 2017 to 62.75 in 2023, indicating a reduction in greenwashing behavior under policy pressure. However, the standard deviation rose sharply to 16.48 between 2021 and 2023, reflecting an increasing disparity in greenwashing behavior among different companies. Some firms responded positively to the policies by improving their environmental performance, while others may have continued or even intensified their greenwashing behaviors, leading to greater divergence in the GWL index.

Since 2020, the issuance of regulations such as the "Measures for the Administration of Legal Disclosure of Enterprise Environmental Information" has further clarified corporate disclosure obligations, resulting in the gradual maturation of China's environmental information disclosure system. Consequently, the average GWL has continued to decline, indicating a general reduction in greenwashing behavior. However, the persistently high maximum value and increased standard deviation still reveal significant disparities in environmental performance among companies. Some firms exhibit high environmental transparency and consistency driven by policy, while others may continue to engage in exaggerated or misleading green marketing practices. This phenomenon suggests that while the maturation of the policy framework has facilitated a reduction in overall greenwashing behavior, inconsistencies among different companies remain pronounced. Future efforts must focus on strengthening policy enforcement and regulation to ensure transparency and authenticity in environmental information disclosures across all firms.

Overall, as China's environmental information disclosure system matures, corporate greenwashing behavior has generally decreased. However, with the deepening of policy implementation, performance disparities among different companies have become increasingly apparent. This indicates that while policies have been effective in curbing greenwashing, further execution and regulation are crucial.

5.3 Multidimensional analysis of corporate greenwashing index

5.3.1 Influence of corporate characteristics

Corporate environmental information disclosure is influenced by multiple factors, among which corporate characteristics are one of the key factors. Existing studies have pointed out that company size, industry classification, and financial conditions all affect the extent of environmental information disclosure in financial reports [51, 52]. Based on this, this study conducts an in-depth analysis of the corporate greenwashing index from the perspective of corporate characteristics, specifically exploring the impact of firm size, industry attributes (whether it belongs to a heavily polluting industry), and financial characteristics on the GWL index.

1. Firm Size (Size)

There is a significantly negative correlation between firm size and GWL, indicating that larger firms have a significantly lower greenwashing index. This result aligns with the resource-based theory, which suggests that large firms usually possess stronger environmental management capabilities and information disclosure standards, and their disclosed content is more likely to reflect true environmental performance [31, 51]. Additionally, large firms, facing higher social supervision pressure and reputation risks, are more inclined to adopt substantive environmental disclosure strategies to reduce greenwashing risks [23, 31]. This finding validates the relationship between firm size and information disclosure quality in relevant studies.

2. Industry Attributes (IND)

There is a significantly negative correlation between industry attributes and GWL, indicating that the greenwashing index of firms in heavily polluting industries is significantly lower than that of firms in non-heavily polluting industries. This is closely related to the differences in the intensity of environmental regulations across industries. Heavily polluting industries face stricter environmental supervision and their disclosure behaviors need to comply with mandatory regulations such as the "Environmental Information Disclosure Guidelines." These mandatory disclosure requirements reduce their room for strategic disclosure. Moreover, heavily polluting industries need to regularly disclose pollutant emission data, leading to more standardized text content, which in turn inhibits strategic disclosure behaviors. Strict environmental regulations may force polluting industries to adopt more cautious disclosure strategies and even reduce greenwashing behaviors due to high compliance costs. In contrast, firms in non-heavily polluting industries have lower costs for managing their environmental image and may exaggerate their environmental contributions through "soft greenwashing" means [47]. At the same time, non-heavily polluting industries have fewer mandatory disclosure requirements, and the transparency of corporate environmental information is generally lower than that of heavily polluting industries, making symbolic disclosure more common [46, 53].

3. Financial Characteristics

Profitability (ROA): Profitability is negatively correlated with GWL, indicating that firms with strong profitability have weaker motives for greenwashing [23]. Firms with strong profitability usually have lower financing needs [23], less external pressure, and are more capable of reducing greenwashing behaviors through transparent governance (such as green certifications and media supervision). Firms with high ROA are likely to reduce symbolic environmental actions (such as greenwashing) and instead improve environmental performance through substantive environmental measures (such as technological improvements and emission reductions) [8]. These firms are more inclined to build competitive advantages through real environmental investments rather than relying on greenwashing behaviors [31]. According to the Signaling Theory, firms with high ROA are more inclined to transmit signals of competitive advantages through real environmental investments and avoid reputation risks caused by greenwashing [8, 54].

Financial Leverage (Lev): Financial leverage is positively correlated with GWL, indicating that firms with high debt have a significantly increased risk of greenwashing. Debt pressure may prompt firms to engage in greenwashing behaviors to obtain green financing or alleviate pressure from stakeholders [23, 47], confirming the positive correlation between financial distress and greenwashing behaviors [23, 48].

5.3.2 Influence of textual features

Most existing studies focus on the relationship between environmental disclosure texts and environmental performance. Some studies suggest that firms with better environmental performance disclose more information [42, 51, 55]. However, other studies have found that poorer environmental performance is associated with more environmental information disclosure [52]. Research has found that non-compliant firms often issue longer, more positive, and more frequent public reports and use these reports to disseminate more environmental content [56].

In view of this, further research on greenwashing should focus on the quality of the environmental texts disclosed by enterprises. Therefore, in this study, the corporate greenwashing behavior and the proportion of effective environmental information disclosed in corporate CSR reports are quantitatively measured. The Normative Disclosure Completeness Ratio (NDCR) of CSR reports is measured by calculating the proportion of substantive actions actually disclosed in the CSR reports to the total environmental information items that should be disclosed in the measuring corporate greenwashing indicator system. This method serves to further elucidate the textual characteristics of enterprises engaged in greenwashing.

1. Normative Disclosure Completeness Ratio (NDCR)

There is a significantly negative correlation between NDCR and GWL, indicating that the higher the degree of greenwashing, the lower the completeness of normative information disclosure in the CSR report, i.e., the fewer information items required by the measuring corporate greenwashing indicator system are disclosed. NDCR reveals the key aspect of corporate greenwashing behaviors: although non-compliant firms may issue longer texts, they do not truly disclose key information. Their disclosure content deviates from the standard framework that should be disclosed, manifesting as insufficient disclosure of key information. The GWL index constructed in this study using a deep learning model can more clearly reveal corporate greenwashing behaviors from the perspective of strategic environmental information disclosure. The GWL index captures the quality of corporate environmental information disclosure through textual structural characteristics and can effectively identify greenwashing behaviors that are “long in form but empty in substance.” The greenwashing indicators constructed from the perspective of textual information structure have more significant implications for measuring the quality of environmental text information disclosure.

Through the analysis of firm-specific attributes and textual characteristics of environmental information disclosure, this study reveals the multidimensional characteristics of the corporate greenwashing index. Firm size, industry attributes, and financial characteristics have significant impacts on the corporate greenwashing index, and the characteristics of environmental information disclosure texts also reveal corporate greenwashing behaviors. These findings provide a multi-perspective understanding of corporate greenwashing behaviors and offer theoretical basis and practical guidance for the regulation and standardization of corporate environmental information disclosure.

6 Conclusion

6.1 Research significance

Firstly, deep learning methods are introduced to construct a corporate greenwashing index, which expands the innovative application paths of different machine learning models in the social sciences. The MacBERT model is capable of processing and analyzing large-scale text data, extracting fine-grained corporate environmental information. Existing studies often employ the dictionary method when constructing indicators. The application of the MacBERT model can take context into account to avoid ambiguous words, enabling a more precise construction of greenwashing indicators at the sentence level, thereby more accurately identifying and quantifying corporate greenwashing behaviors. The application of this method enriches the research methods for corporate information disclosure and environmental behaviors, provides new research perspectives and dimensions, helps to validate and improve existing theories, and lays a solid foundation for the construction of new theories.

Secondly, existing studies have laid a certain foundation for identifying corporate strategic disclosure behaviors, but limitations still exist. For example, simple text size statistics can only provide a single measure based on text length and cannot conduct an in-depth analysis of the authenticity and logicality of the text content. The dictionary method relies on a preset keyword database, making it difficult to cope with strategic disclosures carried out by companies through means such as keyword stacking, and it is easily misled, leading to inaccurate identification results. In contrast, the GWL index based on deep learning can deeply explore the structural and semantic features of texts, accurately identifying greenwashing behaviors that appear compliant in form but are actually hollow in substance. This includes descriptions in corporate reports where environmental-related keywords are deliberately stacked without actual initiatives, as well as statements with intentionally vague logic to cover up the lack of substantial fulfillment of environmental responsibilities. Thus, it provides a more reliable and precise solution for the identification of greenwashing behaviors, effectively addresses the interference of corporate strategic disclosures, and promotes the progress of related research and practices.

Thirdly, the use of machine learning methods overcomes the limitations of traditional approaches regarding data volume, broadening the scope of the study and making the results more representative and generalizable. By mining vast amounts of textual data disclosed by companies, the MacBERT model reveals the deep-seated characteristics and trends of greenwashing behavior. Compared to traditional methods, this model reduces human subjective bias in text analysis, thereby enhancing the objectivity and accuracy of greenwashing identification. Furthermore, the constructed greenwashing index can capture the quality of corporate environmental information disclosure through text structural features, thereby enriching and refining the methods for identifying corporate greenwashing behaviors. The integration of natural language processing (NLP) technology with corporate social responsibility (CSR) research fosters the convergence of management studies, computer science, and social sciences, promoting a diverse and comprehensive development of academic research.

Finally, the application of the MacBERT model not only aids in the accurate identification of corporate greenwashing behaviors but also enhances the transparency and quality of environmental information disclosures, thereby increasing public trust in corporate environmental commitments. The corporate greenwashing index constructed based on this model provides scientific and objective data support for policymakers, assisting in the formulation of more rational and effective environmental policies and regulations, which enhances the scientific and practical aspects of such policies. Additionally, the corporate greenwashing index offers regulatory agencies refined and dynamic corporate environmental data, improving regulatory efficiency and optimizing environmental oversight mechanisms. By constructing the corporate greenwashing index, precise governance of corporate greenwashing behaviors can be achieved, while further strengthening corporate social and environmental responsibilities, encouraging more effective environmental actions, and ultimately improving overall environmental protection efforts, thereby promoting the sustained development of ecological civilization.

6.2 Diverse applications of corporate greenwashing index

The adoption of the MacBERT model to construct corporate greenwashing indicators enriches the methods for identifying corporate greenwashing, introduces new theoretical frameworks and research methodologies to the study of corporate information disclosure and environmental behaviors, and makes significant contributions in terms of theory, methodology, and practical applications. The practical applications of this framework demonstrate broad prospects, offering scientific guidance to stakeholders and solid data support for policy formulation, thereby advancing ecological civilization construction.

The construction of accurate corporate greenwashing identification indicators not only has a wide range of application scenarios but also can provide targeted support for various stakeholders. At the corporate level, the index assists companies in recognizing and mitigating greenwashing behaviors, reducing information asymmetry, and enhancing the transparency of environmental disclosures. Through regular self-assessment and strategic adjustments, firms can optimize their environmental management, lower legal and reputational risks, and strengthen their social image and public trust, thus promoting sustainable development.

In the capital market, ESG rating agencies can utilize the corporate greenwashing index to obtain more accurate data on corporate environmental performance, thereby improving the accuracy and credibility of ratings and supporting investors in making informed investment decisions that encourage green investments. Investors can rely on genuine environmental responsibility metrics to avoid investing in greenwashing companies, thereby minimizing investment risks and achieving sustainable investment goals. Consumers, too, can make more environmentally friendly purchasing choices, steering clear of misleading advertising and fostering a market that leans toward green and sustainable practices.

For regulatory agencies, the greenwashing early warning mechanisms based on the MacBERT model can enhance the proactivity and efficiency of oversight, effectively identifying and preventing corporate greenwashing behaviors and strengthening regulatory scrutiny over corporate environmental performance. The media and the public can leverage this mechanism to gain more accurate insights into corporate environmental performance, thereby reinforcing social oversight and promoting transparency in environmental practices, which can inform policy adjustments.

At the policy formulation level, the quantitative data provided by the corporate greenwashing index can enhance the scientific and rational basis for policymaking. This data feedback can optimize existing policies, increasing transparency and public trust while guiding companies to comply with environmental regulations, ultimately promoting the comprehensive advancement of ecological civilization.

6.3 Comparison and implications

China's "semi-mandatory" policy for corporate social responsibility (CSR) information disclosure is reflected in the imposition of mandatory environmental information disclosure requirements on specific industries, such as heavily polluting enterprises. Meanwhile, China has also introduced market-based incentive mechanisms, such as environmental, social, and governance (ESG) financing preferences, to encourage enterprises to more actively fulfill their social responsibilities and improve their environmental performance. This hybrid model, which combines mandatory policies with market-based incentives, constitutes the core element of China's CSR information disclosure regulatory framework. However, due to the lack of detailed regulatory rules in the initial stages and deviations in the implementation process, enterprises generally tend to adopt symbolic disclosure strategies (e.g., fabricating environmental performance data) rather than substantially promoting green transformation [6, 43, 47]. This behavioral pattern not only weakens the authenticity and effectiveness of environmental information disclosure but also leads to greenwashing behaviors exhibiting distinct instrumental characteristics, where enterprises use political connections and other means to evade regulatory accountability and maintain their own image and interests [7].

Despite the certain lag in the current regulatory system, existing research is actively exploring new paths for constructing a "progressive governance" model in emerging economies. For example, the application of naive Bayes machine learning methods can effectively detect the intangible characteristics of environmental information, thereby improving the precision and efficiency of regulation [6]. Meanwhile, policy experiments are also continuously being promoted, such as the expanded application of environmental taxes, aiming to incentivize enterprises to reduce pollution and improve environmental performance through economic means [5]. In the future, China needs to further advance the localization process of international standards, such as integrating global reporting initiatives (GRI) and other international standards with China's actual situation, to enhance the standardization and comparability of environmental information disclosure. At the same time, the function and role of independent auditing should be strengthened to improve the authenticity and transparency of corporate environmental information [23]. These efforts will lay a solid foundation for China's environmental governance and sustainable development.

This study is not without limitations. Although the sample encompasses listed companies in the A-share market, the effectiveness of the corporate greenwashing index has yet to be validated in other economies. Conducting similar validations in the U.S., EU, and other economic regions would enhance the generalizability of the findings. Additionally, the dataset used in this study is limited to A-share listed companies, with the sample confined to large-scale listed enterprises. Subsequent research could conduct a more detailed analysis focusing on small and medium-sized enterprises to explore the applicability of the existing conclusions across different enterprise sizes. Despite its limitations, the corporate environmental protection indicators constructed in this study using deep learning methods have enhanced data coverage and objectivity, thereby enriching the methods for identifying and evaluating corporate greenwashing behaviors. Future research can replicate this methodology in different economies or expand on various research dimensions, laying the groundwork for further investigations into corporate greenwashing.

Appendix A

Table 4 Measuring corporate greenwashing indicator system

Item			Disclosure status	Substantive actions
1	Governance and Structure	Environmental Policies and Strategies		
2		Environmental Protection Goals and Achievements		
3		Environmental Regulations and Implementation		
4		Environmental Management Institutions and Operations		
5	Processes and Controls	Environmental Certification Systems and Implementation		
6		Environmental Honors and Recognition		
7		Environmental Investment and Comprehensive Remediation Plans		
8		Environmental Education, Training, and Public Welfare Activities		
9		Environmental Technology Development and Innovation		
10	Inputs and Outputs	Energy Consumption and Reduction Measures		
11		Water Resource Consumption and Reduction Measures		
12		Greenhouse Gas Emissions and Reduction Measures		
13		Air Emissions and Reduction Measures		
14		Wastewater Generation and Reduction Measures		
15		Solid Waste Generation and Disposal Measures		
16		Additional Emission Reduction Measures (Greening, Noise Control, and Logistics)		
17	Compliance and Legality	Compliance Statements Regarding Environmental Laws and Regulations		
18		Risk Assessment Associated with Environmental Policies		
19		Impact of Industry Characteristics on Environmental Effects		
20		Disclosure of Major Environmental Pollution Incidents		

Appendix B

Table 5 Examples of labeled samples

Year	Code	Substantive action	Sentences
2022	000002	0	In 2022, 20 shopping malls under Yinli Commercial achieved a reduction in greenhouse gas emissions through photovoltaic power generation.
2009	000157	1	Obtained ISO14001 Environmental Management System certification in 2006 and passed recertification audit in 2009.
2010	000069	0	Paper packaging enterprises actively introduce energy-saving equipment and devices, while upgrading or phasing out inefficient and high-energy-consuming machinery to reduce energy consumption.
2010	000825	1	The company's annual secondary energy recovery accounts for 48% of total energy consumption, with waste heat and pressure power generation covering 24% of total electricity usage.
2010	000060	1	From January to October 2010, Shaoguan Smelter saw a 9.8% year-on-year decrease in monthly average exhaust emissions, including a 5.7% reduction in SO2 emissions.
2012	000550	1	Invested ¥250,000 in frequency conversion retrofitting for 2 air supply units at the coating line II of the Body Plant, achieving annual savings of ¥137,000 and conserving 18.8 metric tons of standard coal equivalent.
2012	000630	1	Precious metals investment implemented energy-saving upgrades including cyclonic electrodes, furnace lance systems, and motor frequency conversion, realizing annual electricity savings of 5.56 million kWh.
2012	000630	1	In recent years, the company has invested over ¥10 million in comprehensive energy conservation and environmental protection initiatives.

Table 5 (*Continued*)

Year	Code	Substantive action	Sentences
2020	000530	0	The company adheres to process energy conservation, product efficiency enhancement, efficient energy management, and a continuous improvement environmental policy.
2020	000050	1	Completed projects are projected to annually reduce energy consumption by approximately 2000 metric tons of standard coal equivalent and decrease wastewater discharge by over 1 million metric tons.
2020	000055	0	Promoting green office practices by reducing standby power consumption of air conditioners and computers, while optimizing workspace temperature settings for energy efficiency.
2018	000090	0	In 2018, the company achieved notable results through technological innovation and application of new construction technologies, processes, and methodologies.
2018	000598	1	2017 R&D investment totaled ¥9.5069 million, initiating multiple water-related and environmental protection research projects.
2018	000701	0	Xinda Optoelectronics IoT Research Institute drives eco-friendly development in electronic information through new materials and low-cost environmental innovations.
2015	000069	1	In 2015, Taizhou OCT reduced natural gas consumption by 127,000 m^3, electricity use by 740,000 kWh, and water usage by 40,000 m^3.
2015	000778	1	Strictly complies with the National Emergency Response Plan for Environmental Incidents, with zero pollution incidents occurring in 2015.
2013	000725	1	Water resource management: Hefei Sixth Generation Line's 2013 COD discharge decreased by 1.66 tons to 7.88 tons year-on-year, with ammonia nitrogen reductions.
2013	000937	0	Gequan Mine reduces resource consumption through material reuse and production cost control.
2013	002162	1	Post-industrial restructuring, energy consumption decreased by 21,244 metric tons of standard coal equivalent, reducing carbon emissions by 42,485 metric tons.
2013	000877	0	Implemented oxygen-enriched combustion technology in 2013 for enhanced energy conservation and emission reduction.
2013	000877	0	Continuous adoption of new technologies drives the company's sustainable development toward low-carbon environmental protection.
2012	000709	1	2012 industrial water recycling rate exceeded 97%, with recovered energy constituting two-thirds of total consumption.

Abbreviations
GWLS, Selective Disclosure; GWLE, Expressive Manipulation; GWL, Degree of Corporate Greenwashing; CSR, Corporate Social Responsibility; ESG, Environmental, Social, and Governance; PRI, Principles for Responsible Investment; NLP, Natural Language Processing; NSP, Next Sentence Prediction; MLM, Masked Language Modeling; NDCR, Normative Disclosure Completeness Ratio.

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Author contributions
XW: Formal analysis, investigation, methodology, software development, data labeling, writing—original draft. XG: Conceptualization, data labeling, investigation. MS: Data labeling, writing—review & editing.

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Data availability
Data will be made available on reasonable request.

Declarations

Ethics approval and consent to participate
This study was conducted in accordance with ethical standards. As this research primarily involves text mining and machine learning techniques to analyze publicly available corporate environmental disclosures, it does not require ethical approval from a formal ethics committee. No human participants were involved in this study.

Consent for publication
The authors hereby affirm their consent to the publication of this manuscript in the journal. Furthermore, they confirm that all relevant data, methodologies, and results have been fully disclosed and presented in the manuscript.

Competing interests
The authors declare no competing interests.

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